Project Dataset Understanding

Oct 16, 2026 · @Saurabh

1. Dataset overview

The project dataset is a **synthetic (fictional) teaching dataset** of 1,100,000 Indian digital payments (1,000,000 main + 100,000 warm-up), with fraud labels, pre-computed features, simulated decisions, investigator text and LLM training files. It exists to build and test the project real-time fraud scorer described in the CRD, HLD and LLD. Generator version 1.0.1, seed 20261010, label snapshot 1 May 2026.

Business problem: decide in real time whether each payment should be ALLOWED, sent to REVIEW or BLOCKED, and explain why, while catching rare fraud (0.6% of payments) without annoying genuine customers.

What one record represents: one payment authorisation attempt by a customer, at a merchant, from a device, at a moment in time, plus everything learned about it (its outcome, its fraud label 60 days later, its features, the system's decision).

Size and structure

The folder is about 750 MB: 18 relational tables (as gzipped CSV, plus Parquet for transactions), JSONL files for histories and LLM training, and Markdown documents. Most tables are split by time into train, validation and holdout; only transactions, payment_outcomes, labels and features also have a warm-up part (Oct–Dec 2024 payments that give customers history before January 2025), and reference tables like customers and merchants are not split.

Area (folder)	Contents	Records	Format
`tables/`	18 relational tables (customers, transactions, features, labels, decisions, rules, text metadata)	see below	CSV.gz
`parquet/transactions/`	Same transactions joined with their labels: 19 columns (no merchant_id; adds fraud_label and label_available_at)	1,100,000	Parquet (52 files)
`histories/`	Last ≤ 20 prior payments per transaction (what Redis would hold)	one per transaction	JSONL.gz
`training/`	SFT prompts, CPT corpus, DPO pairs, gold explanations, fmt-v1 vs fmt-v2 comparison	714,745 SFT train rows; 5,000 DPO; 8,000 gold	JSONL.gz
`corpus/`	Investigator notes, chargeback narratives, playbooks, synthetic RBI	39,006	JSONL.gz

Area (folder)	Contents	Records	Format
	exercises, guidelines	documents	
`documents/`	Human-readable Markdown copies of a sample of the corpus	1,525 files	.md
`ground_truth/`	Generator scenario per event (evaluation only, never a model input)	1,100,000	JSONL.gz
`evaluation/`	500-case holdout explanation pack	500	JSONL.gz
`sql/`	Postgres schema, indexes, validation queries, training-role example	4 scripts	.sql
Root files	Data card, data dictionary, feature catalogue, manifest, quality report, rules config, requirement coverage	10 files	.md / .json

Record counts per table

Table	Rows	Columns	Split
transactions	1,100,000	18	warm-up 100,000 · train 714,745 · validation 144,803 · holdout 140,452
payment_outcomes	1,100,000	5	same as transactions
labels	1,100,000	7	same as transactions
features	1,100,000	22	same as transactions
decision_audit	1,000,000	18	main splits only
devices	100,120	3	all
document_chunks	78,007	8	train / validation / holdout
documents	39,006	16	train / validation / holdout
customers	25,000	6	all
reviewer_feedback	19,376	7	train / validation / holdout
gold_explanations	8,500	8	train 8,000 · holdout 500
preference_pairs	5,000	9	train
merchants	2,000	5	all
legacy_cases	2,000	9	all (2021–2024 archive)

Table	Rows	Columns	Split
list_entries	137	6	all
hard_rules	2	7	all
threshold_sets	1	8	all
config_changes	1	9	all

Across the 18 tables there are **171 attributes**. Data types present: text IDs, integers, decimals, UTC timestamps, Booleans, JSON objects (locations, reason codes, request snapshots) and free text.

Time splits (IST)

Split	Period	Rows	Fraud	Fraud rate
Warm-up	Oct–Dec 2024	100,000	600	0.60%
Train	Jan–Oct 2025	714,745	4,236	0.59%
Validation	Nov–Dec 2025	144,803	928	0.64%
Holdout	Jan–Feb 2026	140,452	836	0.60%

How the tables relate

`transactions` is the centre. Each transaction points to one **customer**, one **merchant** and (99% of the time) one **device**. Four tables share its `transaction_id` one-to-one: **payment_outcomes**, **labels**, **features** and **decision_audit**. Text and training tables (documents, gold explanations, preference pairs, reviewer feedback, SFT prompts, histories) also link back by `transaction_id`. Rules tables (threshold_sets, hard_rules, list_entries, config_changes) stand alone as configuration. Every foreign key checked resolves: 0 orphan customer, merchant or device IDs.

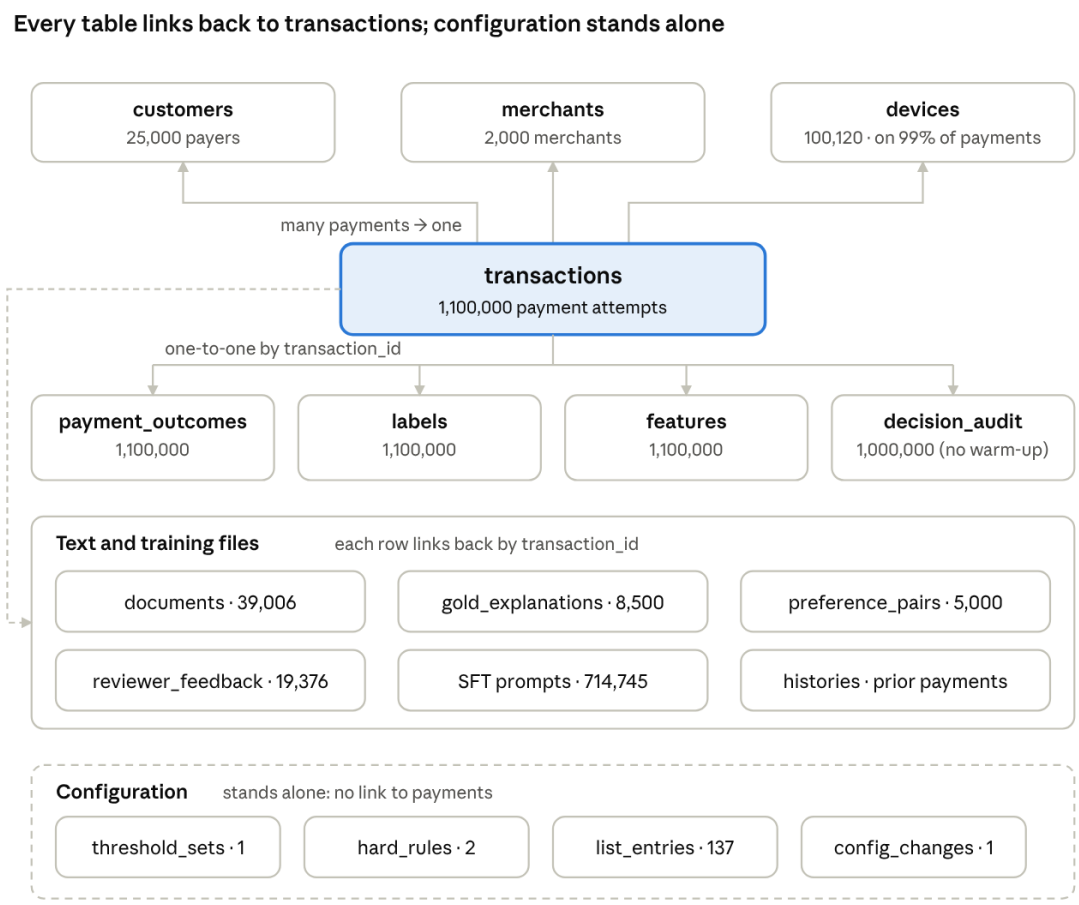


table relationships · data_dictionary.json, manifest.json and foreign-key checks

Read it top to bottom: many payments point up to one customer, merchant and device; four tables sit one-to-one under each payment; text and training files link back by `transaction_id`; the dashed configuration box has no link to payments at all.

2. Data dictionary

Every attribute of every table is listed below with its type, business meaning, observed values and its role in project. Types are the Postgres types from `data_dictionary.json`; values are measured from the full data. `is_synthetic` appears in every table and is always `true` (a flag marking the row as fictional); it is listed once here and not repeated.

Under each table is **one real example row** taken from the files. Where tables link, the examples follow the same payment, **TX_0000100016**: a ₹1,418.66 wallet payment for electronics at 00:10 IST on 1 Jan 2025. It was later confirmed as fraud, but the simulated system allowed it. Timestamps are stored in UTC; the IST time is shown in brackets.

2.1 customers (25,000 rows) — one row per payer

What it is: One row per person who pays: when they opened their account, their age, home city and region. Payments link here by customer_id, so you can ask whether a payment looks normal for this person.

Attribute	Type	Business meaning	Observed values	Use in project
customer_id	text, PK	Pseudonymised customer code	C_ + 16 hex chars; 25,000 unique	Joins to transactions; Redis key for history and velocity
created_at	timestamptz	When the account was opened	1 Jan 2020 to 27 Feb 2026 (IST)	Source of account age (FE-06)
customer_age	integer	Customer age in years	18–80, median 49	Model input; fairness checks by age band
home_city	text	Customer's home city	16 cities, ~1,550 each (Kochi, Delhi, Lucknow…)	Baseline for new-city / geo checks
region	text	Indian region	east 6,261 · south 6,236 · north 4,765 · west 4,669 · northeast 1,541 · central 1,528	Fairness checks by region (FR-ML-11)
is_synthetic	boolean	Row is fictional	always true	Data-governance marker

One example row (the payer of TX_0000100016)

Column	Example value	What it tells you
customer_id	C_625a6f3e64f87d0f	Pseudonymous ID for this payer
created_at	2023-06-29T18:30:00Z (30 Jun 2023, 00:00 IST)	Account opened 551 days before the payment
customer_age	62	Age in years
home_city	Guwahati	Usual city of the payer
region	northeast	Region of the home city
is_synthetic	true	Fictional record

2.2 merchants (2,000 rows) — one row per merchant

What it is: One row per shop or service that receives money, with its category (for example electronics or gaming) and city. The category matters most: gift cards, gaming and electronics hold 97% of the fraud.

Attribute	Type	Business meaning	Observed values	Use in project
merchant_id	text, PK	Merchant code	`M_` + 16 hex; 2,000 unique	Joins to transactions; allowlist target
synthetic_name	text	Fictional merchant name	"Synthetic Merchant 000000"…	Display only; not a model input
merchant_category	text	Merchant category (MCC group)	10 groups, 200 merchants each: grocery, food, education, travel, electronics, gaming, gift_cards, quick_commerce, healthcare, other	Strong risk signal; per-MCC thresholds (FR-RE-04); FE-08
merchant_location	JSON	Merchant city, latitude, longitude	16 cities, 125 merchants each	Distance and geo-mismatch features (FE-03)

One example row (the merchant paid in TX_0000100016)

Column	Example value	What it tells you
merchant_id	M_aad2c9114ea977bf	Merchant code
synthetic_name	Synthetic Merchant 000154	Placeholder name, no real brand
merchant_category	electronics	One of the three high-fraud categories
merchant_location	{"city":"Kochi","lat":9.93,"lon":76.27}	City and coordinates of the merchant
is_synthetic	true	Fictional record

2.3 devices (100,120 rows) — one row per device fingerprint

What it is: One row per phone or browser used to pay, saying whether it is a mobile app or a web browser. It lets you spot a device that is new for a customer, or one device shared by many customers, which is a strong fraud signal.

Attribute	Type	Business meaning	Observed values	Use in project
device_id	text, PK	Device fingerprint code	`D_` + 16 hex; 100,120 unique (53,018 used in transactions)	New-device (FE-02), device sharing (FE-09), blocklist
device_kind	text	Type of device	mobile 75,090 · browser 25,030	Possible model input / segment

One example row (the phone used for TX_0000100016)

Column	Example value	What it tells you
device_id	D_3cdd123ac0cca915	Device fingerprint
device_kind	mobile	A phone app, not a browser
is_synthetic	true	Fictional record

2.4 transactions (1,100,000 rows) — one row per payment attempt

What it is: One row per payment attempt: who paid whom, how much, when, from where, by which channel and on which device. It is the central table; every other payment table joins to it by transaction_id.

This is the core table: the request the Payment API sends (CRD §10.2) plus the historical fields the feature store supplies.

Attribute	Type	Business meaning	Observed values	Use in project
transaction_id	text, PK	Unique payment ID	`TX_` + 10 digits; 1,100,000 unique, 0 duplicates	Join key to every other table; audit ID
customer_id	text, FK	Who paid	24,834 distinct customers; busiest has 109 payments	Lookup of history, velocity, allowlist
merchant_id	text, FK	Who was paid	2,000 distinct; busiest 1,161 payments	Merchant risk, allowlist
device_id	text, FK, nullable	Device used	53,018 distinct; **10,683 null (0.97%)**	New device, device sharing, blocklist; null → `unk`
amount_paise	bigint	Amount in paise (authoritative)	217 to 798,733,717 (₹2.17 to ₹79.9	Amount and amount-spike

Attribute	Type	Business meaning	Observed values	Use in project
			lakh); median ₹1,250.64	features; integer avoids rounding errors
transaction_amount	numeric(14,2)	Same amount in rupees	always equals amount_paise ÷ 100 (0 mismatches)	API-edge field; redundant with amount_paise
payment_channel	text	How the payment was made	UPI 637,591 · card 341,229 · wallet 76,510 · net_banking 44,670	Model input; weak fraud signal
merchant_category	text	Merchant group (copied from merchant)	10 groups; grocery largest (203,387)	Strongest single fraud signal; per-MCC thresholds
transaction_time	timestamptz (UTC)	When the payment was attempted	1 Oct 2024 – 28 Feb 2026 IST; unique per row	Hour, night flag, split assignment, velocity windows
customer_age	integer	Customer age at payment	18–80, median 49; matches customers table 100%	Model input; fairness
customer_location	JSON, nullable	Customer city + lat/lon	16 cities; **33,067 null (3.0%)**	Distance and new-city features; null → `unk`
merchant_location	JSON	Merchant city + lat/lon	16 cities, never null	Distance feature
account_age	integer (days)	Days since account opened	0–2,249, median 754	New-account flag (< 30 days)
previous_transactions	integer	Customer's prior payments in the last 30 days (feature-store field)	0–35, median 3	History depth; cold-start detection
average_transaction_value	numeric, nullable	Customer's average amount over the preceding 90	₹12.97 to ₹76.8 lakh; **28,325 null (2.6%) = no history**	Amount-to-average ratio (FE-01)

Attribute	Type	Business meaning	Observed values	Use in project
		days, in rupees		
failed_transactions	integer	Failed attempts in the preceding 24 hours	0–9; 0 for 85% of rows	Failure features (FE-05); strong signal
split	text	Which time split the row belongs to	warm-up / train / validation / holdout	Prevents leakage; never a model input

One example row (TX_0000100016)

Column	Example value	What it tells you
transaction_id	TX_0000100016	This payment
customer_id	C_625a6f3e64f87d0f	Who paid (row in customers)
merchant_id	M_aad2c9114ea977bf	Who was paid (row in merchants)
device_id	D_3cdd123ac0cca915	Which phone was used (row in devices)
amount_paise	141866	₹1,418.66 in paise
transaction_amount	1418.66	Same amount in rupees
payment_channel	wallet	Paid from a digital wallet
merchant_category	electronics	Copied from the merchant
transaction_time	2024-12-31T18:40:47.105Z	1 Jan 2025, 00:10 IST: just after midnight
customer_age	62	Copied from the customer
customer_location	{"city":"Chennai","lat":13.08,"lon":80.27}	Paid from Chennai, not the home city Guwahati
merchant_location	{"city":"Kochi","lat":9.93,"lon":76.27}	Merchant is in Kochi, 559 km away
account_age	551	Account is 551 days old
previous_transactions	3	Three earlier payments in the last 30 days
average_transaction_value	628.52	Average past payment ₹628.52

Column	Example value	What it tells you
failed_transactions	1	One failed attempt in the last 24 hours
split	train	Falls in the training period
is_synthetic	true	Fictional record

2.5 features (1,100,000 rows) — the 10 engineered features FE-01 to FE-10 per payment

What it is: For each payment, ready-made signals calculated from the customer's past, such as "2.3 times the usual amount", "new device" or "paid at night". These are what a model actually reads, and they decide which reason codes fire.

All features are computed **at authorisation, from history before the current payment** (point-in-time), so they are safe to use for training.

Attribute	Type	Business meaning	Observed values	Use in project
transaction_id	text, PK/FK	Payment the features belong to	1:1 with transactions	Join key
feature_as_of	timestamptz	Moment the features were computed (= transaction time)	unique per row	Proves no future data used
amount_to_avg_ratio	double, nullable	FE-01: amount ÷ 90-day average	0–32,318, median 0.75; 28,325 null (no history)	Amount-spike signal (R05)
is_new_device	boolean, nullable	FE-02: device not seen for this customer in 90 days	true 72,418 · false 1,016,899 · null 10,683	New-device signal (R01)
customer_merchant_distance_km	double, nullable	FE-03: straight-line (haversine) distance	0–2,429 km, median 1,082; 33,067 null	Geo signal (R02)
is_new_city	boolean, nullable	FE-03: city not seen in 90 days	true 76,679 · null 33,067	Geo signal
hour_of_day	integer	FE-04: IST hour	0–23, **hour 5 never occurs**	Time-of-day signal
is_night	boolean	FE-04: 00:00–05:00 IST	true 155,633 (14%)	Odd-hour signal (R03)
has_recent_failures	boolean	FE-05: ≥ 3 failed attempts in 24 h	true 98,634 (9%)	Failure signal (R04)

Attribute	Type	Business meaning	Observed values	Use in project
is_new_account	boolean	FE-06: account younger than 30 days	true 11,541 (1%)	New-account signal (R06)
txn_count_last_1h	integer	FE-07: prior attempts in the last hour (declines included)	0–17; 0 for 57%	Velocity signal (R07)
txn_amount_last_24h_paise	bigint	FE-07: amount attempted in the last 24 h	0 to ₹80 lakh; median 0	Velocity signal
distinct_customers_per_device_7d	integer, nullable	FE-09: other customers seen on this device in 7 days	0–8; 10,683 null	Device-sharing / mule signal
combined_risk_flags	integer	FE-10: count of device, geo, night, failure and new-account flags	0–4; 1 for 65%	Simple combined risk score
mcc_target_encoded_rate	double	FE-08: smoothed past fraud rate of the merchant category	0.00003–0.027	Category risk without leaking future labels
mcc_encoding_as_of	timestamptz	Month-end date of the labels used for FE-08	14 month-ends; frozen at 1 Nov 2025 IST for validation/holdout	Leakage control
mcc_encoding_count	integer	Number of labelled payments behind FE-08	0–105,167	Confidence in FE-08
missing_customer_location	boolean	Location was absent	true 33,067	Explicit missing flag
missing_device	boolean	Device was absent	true 10,683	Explicit missing flag
cold_start	boolean	Customer had no 90-day history	true 28,325	Explains null FE-01
reason_codes	JSON list	Rule-derived weak reason labels	0–7 codes per row; `[]` for 120,818 rows	Training target for the 8 reason heads

Attribute	Type	Business meaning	Observed values	Use in project
feature_version	text	Version of the feature code	always `project-features-1.0.0`	Reproducibility, training-serving parity

One example row (features of TX_0000100016)

Column	Example value	What it tells you
transaction_id	TX_0000100016	The payment these features describe
feature_as_of	2024-12-31T18:40:47.105Z	Computed at the moment of the payment, using only earlier data
amount_to_avg_ratio	2.2571	₹1,418.66 ÷ ₹628.52: about 2.3 times the usual amount
is_new_device	false	This phone was seen 42 seconds earlier
customer_merchant_distance_km	559.09	Chennai to Kochi
is_new_city	false	Chennai was already in the history
hour_of_day	0	Midnight hour in IST
is_night	true	Between 00:00 and 05:00 IST
has_recent_failures	false	Only 1 failure in 24 hours; the flag needs 3
is_new_account	false	551 days old, not under 30
txn_count_last_1h	1	One payment in the previous hour (TX_0000100013)
txn_amount_last_24h_paise	36858	₹368.58 spent in the previous 24 hours
distinct_customers_per_device_7d	1	Only this customer used the phone
combined_risk_flags	2	Two risk flags raised together
mcc_target_encoded_rate	0.006	Historical fraud rate used for electronics

Column	Example value	What it tells you
mcc_encoding_as_of	2024-12-31T18:30:00Z	Rate frozen at 1 Jan 2025, 00:00 IST
mcc_encoding_count	0	No mature labels behind the rate yet; 0.006 equals the overall 0.6% fraud rate
missing_customer_location	false	Location present
missing_device	false	Device present
cold_start	false	The customer has history
reason_codes	["R02_GEO_MISMATCH","R03_ODD_HOUR","R08_HIGH_RISK_MCC"]	Three reason codes fired (see 2.6)
feature_version	project-features-1.0.0	Version of the feature code

2.6 Reason codes R01–R08

What it is: Not a file of its own, but a fixed list of eight short labels that say why a payment looks risky, such as R03_ODD_HOUR. Each payment carries the codes that fired, and they feed the explanation shown to analysts.

The empty `reason code.docx` in the root can be filled from this table. Names come from the data; meanings from the explanation templates.

Code	Meaning (template wording)	Rows flagged (features)
R01_NEW_DEVICE	Device not observed for this customer in the preceding 90 days	72,418
R02_GEO_MISMATCH	Unusual observed city or customer-to-merchant distance above 500 km	895,465
R03_ODD_HOUR	Authorisation requested between midnight and 05:00 IST	155,633
R04_FAILED_ATTEMPTS	At least three failed payment outcomes in the preceding 24 hours	98,634
R05_AMOUNT_SPIKE	Amount is at least five times the observed 90-day average	30,690
R06_NEW_ACCOUNT	Account age is below 30 days	11,541
R07_VELOCITY	At least four prior authorisations in	148,480

Code	Meaning (template wording)	Rows flagged (features)
	the preceding hour	
R08_HIGH_RISK_MCC	Elevated category flag under the synthetic teaching policy	324,094

One example (which codes fired for TX_0000100016)

Code	Fired?	Why
R01_NEW_DEVICE	No	The phone was used 42 seconds earlier
R02_GEO_MISMATCH	Yes	Customer to merchant distance 559 km, above 500
R03_ODD_HOUR	Yes	00:10 IST is between midnight and 05:00
R04_FAILED_ATTEMPTS	No	1 failure in 24 hours; needs 3
R05_AMOUNT_SPIKE	No	2.3 times the average; needs 5
R06_NEW_ACCOUNT	No	551 days old; needs under 30
R07_VELOCITY	No	1 payment in the past hour; needs 4
R08_HIGH_RISK_MCC	Yes	Electronics is on the high-risk category list

2.7 labels (1,100,000 rows) — the fraud answer for each payment

What it is: The answer key: for each payment, whether it turned out to be fraud, how that was confirmed and the date the answer became known. The model learns to predict fraud_label; the date stops it learning from answers that were not yet known.

Attribute	Type	Business meaning	Observed values	Use in project
transaction_id	text, PK/FK	Payment being labelled	1:1 with transactions	Join key
fraud_label	smallint (0/1)	Was it fraud?	1 = 6,600 (0.6%) · 0 = 1,093,400	**The prediction target**
label_source	text	How the label was decided	mature_no_fraud 1,093,400 · investigator_verdict 5,185 · confirmed_chargeback 1,415	Label quality; chargebacks are strongest evidence
label_available_at	timestamptz	When the label became known	always exactly 60 days after the payment	Label-maturity rule; time-split

Attribute	Type	Business meaning	Observed values	Use in project
				training
split	text	Time split	as transactions	Leakage control
eligible_at_next_split_start	boolean	Was the label already known when the next split began?	true 814,860 · false 285,140 (train 80% true, validation 2% true)	Strict historical simulation; calibration choice

One example row (label of TX_0000100016)

Column	Example value	What it tells you
transaction_id	TX_0000100016	The payment being labelled
fraud_label	1	It was fraud
label_source	investigator_verdict	A fraud analyst confirmed it
label_available_at	2025-03-01T18:40:47.105Z	Answer known on 2 Mar 2025 (IST), 60 days after the payment
split	train	Training period
eligible_at_next_split_start	true	Already known when validation began on 1 Nov 2025, so it can be used for training
is_synthetic	true	Fictional record

2.8 payment_outcomes (1,100,000 rows) — what happened to each payment

What it is: What happened after each payment was sent for authorisation: success or failure, and how quickly. Failed attempts feed the "recent failures" signal (R04).

Attribute	Type	Business meaning	Observed values	Use in project
transaction_id	text, PK/FK	Payment	1:1 with transactions	Join key
outcome_time	timestamptz	When the outcome arrived	0.05–2 s after the payment	Feature updater ordering (only past outcomes count)
status	text	Payment result	SUCCESS 981,435 · FAILED 118,565 (10.8%)	Feeds failed-attempt counters (FE-05)
failure_code	text, nullable	Why it	`SIMULATED_AUTH_FAILURE` when FAILED, else	Failure detail; null is

Attribute	Type	Business meaning	Observed values	Use in project
		failed	null (89%)	expected, not missing

One example row (outcome of TX_0000100016)

Column	Example value	What it tells you
transaction_id	TX_0000100016	The payment
outcome_time	2024-12-31T18:40:48.384Z	Result came back 1.3 seconds after the request
status	FAILED	The payment did not go through
failure_code	SIMULATED_AUTH_FAILURE	Generic simulated authorisation failure
is_synthetic	true	Fictional record

2.9 ground_truth (1,100,000 JSONL records) — generator scenario

What it is: The generator's private notes on which scenario produced each event: normal behaviour or a sudden change of context. It is only for checking how the dataset was made and must never be given to a model.

Field	Type	Meaning	Observed values	Use in project
event_number	integer	Event sequence number	1–1,100,000	Join to the event
scenario	text	Pattern the simulator generated	routine_or_legitimate_variation 923,633 · context_shift_burst 176,367	**Evaluation and analysis only — never a model input**
split	text	Time split	as transactions	—
use	text	Usage rule	always `generator_evaluation_only_never_model_input`	Governance reminder

One example record (the first line of the file; records hold no transaction ID)

Field	Example value	What it tells you
event_number	1	First event the generator produced
scenario	routine_or_legitimate_variation	The generator meant this as normal behaviour
split	warmup	Warm-up period
use	generator_evaluation_only_nev	Only for checking the generator; never a model

Field	Example value	What it tells you
	er_model_input	input

2.10 decision_audit (1,000,000 rows) — a simulated record of every decision

What it is: A log of what the fraud system decided for each main payment: the score, risk level, ALLOW / REVIEW / BLOCK, reasons, explanation, timings and the exact inputs used. It shows what a real audit record looks like and gives a baseline to beat, since it let 89% of fraud through.

This mirrors the 5-year audit record in LLD §3.3. **All scores and decisions here come from a simple heuristic labelled `SYNTHETIC-HEURISTIC-NOT-A-MODEL-v1`, not a trained model.**

Attribute	Type	Business meaning	Observed values	Use in project
transaction_id	text, PK/FK	Payment decided	1:1 with main transactions	Join key
scored_at	timestamptz	When it was scored	= transaction time	Audit trail
fraud_probability	numeric(7,6)	Simulated fraud score	0.0074–0.88, median 0.021	Example of the API field; **not calibrated, never a feature**
decision	text	ALLOW / REVIEW / BLOCK	ALLOW 981,690 · REVIEW 11,644 · BLOCK 6,666	Shows the decision mix (MLOPS-01)
risk_level	text	Risk band	LOW 937,304 · MEDIUM 50,489 · HIGH 12,180 · CRITICAL 27	Shows band → decision mapping
reason_codes	JSON list	Reasons returned	up to 3 codes	Example of API output
explanation	text	Plain-English reason	85 template sentences	Template fallback examples
decision_source	text	Who decided	MODEL 993,360 · HARD_RULE 6,640	FR-RE-05 example
model_version	text	Model that scored it	always the heuristic label	Audit field
rules_version	text	Rules active	always `synthetic-`	Audit field

Attribute	Type	Business meaning	Observed values	Use in project
			`rules-1`	
latency_ms	integer	Simulated decision time	20–220 ms, median 120	Example only; not real performance
explanation_latency_ms	integer	Simulated explanation time	20–1,720 ms, median 870	Example only
prompt_format	text	Prompt format used	always fmt-v1	Audit field
prompt_hash	text	SHA-256 fingerprint of the prompt	64 hex chars	Prove which prompt was used without storing it
request	JSON	Request exactly as received	CRD §10.2 fields	Replay and audit
feature_vector	JSON	Feature values used	FE-01 to FE-10	Replay and audit
split	text	Time split	train / validation / holdout	—

One example row (decision for TX_0000100016: a fraud payment the heuristic missed)

Column	Example value	What it tells you
transaction_id	TX_0000100016	The payment scored
scored_at	2024-12-31T18:40:47.105Z	Scored at the payment time
fraud_probability	0.066175	About 7% fraud risk, below the 0.30 MEDIUM cut-off
decision	ALLOW	Let through, although it was fraud
risk_level	LOW	Band for a score under 0.30
reason_codes	["R02_GEO_MISMATCH","R03_ODD_HOUR","R08_HIGH_RISK_MCC"]	Codes shown with the decision
explanation	"Unusual observed city or customer to merchant distance above 500 km; Authorisation requested between midnight and 05:00 IST; Elevated category flag …"	Template text built from the codes

Column	Example value	What it tells you
decision_source	MODEL	Decided by the score, not a hard rule
model_version	SYNTHETIC-HEURISTIC-NOT-A-MODEL-v1	A simple heuristic, not a trained model
rules_version	synthetic-rules-1	Rules in force
latency_ms	41	Scoring took 41 ms
explanation_latency_ms	746	Explanation took 746 ms
prompt_format	fmt-v1	Prompt layout used
prompt_hash	10be16a0…3eca (64 hex characters)	Fingerprint of the exact prompt
request	{"transaction_id":"TX_0000100016","transaction_amount":1418.66,"merchant_category":"electronics",…}	The 10 request fields the API received
feature_vector	{"amount_to_avg_ratio":2.2571,"is_night":true,…}	The features used, copied from 2.5
split	train	Training period
is_synthetic	true	Fictional record

2.11 Rules and configuration tables

What it is: Settings, not payments: the score cut-offs for each risk level, hard rules such as "always block a blocklisted device", the block and allow lists, and a log of who changed what. The rules engine reads these to turn a score into a decision.

Table.attribute	Type	Meaning	Observed values
threshold_sets.threshold_id	text, PK	Threshold set name	`THR_DEFAULT_V1`
threshold_sets.version	integer	Version number	1
threshold_sets.mcc_group	text, nullable	Merchant group it applies to	null = default for all
threshold_sets.medium_from / high_from / critical_from	numeric(4,3)	Probability cut-offs for MEDIUM / HIGH / CRITICAL	0.30 / 0.60 / 0.85
threshold_sets.status	text	Lifecycle state	`active_synthetic`

Table.attribute	Type	Meaning	Observed values
hard_rules.rule_id / code	text	Rule name	HR_BLOCKLIST_DEVICE, HR_ALLOWLIST_CUSTOMER
hard_rules.expression	JSON	Safe rule condition	`device_id in_list blocked_devices; customer_id in_list allowlisted_customers`
hard_rules.action	text	What the rule does	BLOCK; CAP_AT_REVIEW
hard_rules.priority	integer	Which rule wins	100 (block) beats 10 (allowlist)
hard_rules.enabled	boolean	On/off	both on
list_entries.entry_id	text, PK	List row ID	137 entries
list_entries.list_type	text	Which list	blocked_devices 12 · allowlisted_customers 125
list_entries.entity_id	text	Device or customer on the list	`D_…` / `C_…`
list_entries.effective_from	timestamptz	Start date	1 Jan 2025 IST
list_entries.reason	text	Why listed	"fictional seeded policy fixture"
config_changes.change_id	text, PK	Change ID	CHANGE_0001
config_changes.entity / entity_id	text	What was changed	threshold_sets / THR_DEFAULT_V1
config_changes.before_value / after_value	JSON	Values before and after	null → {medium 0.3, high 0.6, critical 0.85}
config_changes.changed_by / changed_at / reason	text, timestamptz, text	Who, when, why	SYNTHETIC_ADMIN, 1 Jan 2025 IST, "Initial CRD teaching thresholds"

One example row from each table

Table	Example row	What it tells you
threshold_sets	THR_DEFAULT_V1 · version 1 · mcc_group empty · medium_from 0.300 · high_from 0.600 · critical_from 0.850 · active_synthetic	Default cut-offs for every category: 0.30 MEDIUM, 0.60 HIGH, 0.85 CRITICAL
hard_rules	HR_BLOCKLIST_DEVICE · {"field":"device_id","operator":"in_list","list":"block	Block any payment from a device on the blocklist,

Table	Example row	What it tells you
	ed_devices"} · BLOCK · priority 100 · enabled true	whatever the score
list_entries	blocked_devices_0 · blocked_devices · D_03d90b6733788ce8 · from 2024-12-31T18:30:00Z · "fictional seeded policy fixture"	Puts this device on the blocklist from 1 Jan 2025 (IST)
config_changes	CHANGE_0001 · threshold_sets / THR_DEFAULT_V1 · before empty · after {"medium":0.3,"high":0.6,"critical":0.85} · SYNTHETIC_ADMIN · 2024-12-31T18:30:00Z · "Initial CRD teaching thresholds"	Audit trail: who set the thresholds, when and why

`rules_config.json` repeats the same lists and thresholds (named medium 0.3, review 0.6, block 0.85), states that a device block beats a customer allowlist, and that scores are not calibrated.

2.12 reviewer_feedback (19,376 rows) — investigator verdicts on REVIEW cases (MLOPS-08)

What it is: For payments sent to REVIEW, what the human analyst concluded (FRAUD or GENUINE) and how they would reword the explanation. It is the feedback loop for retraining the scorer and improving explanations.

Attribute	Type	Business meaning	Observed values	Use in project
feedback_id	text, PK	Feedback ID	`FB_` + transaction ID	Join key
transaction_id	text, FK	Case reviewed	19,376 distinct	Link to decision
reviewed_at	timestamptz	When reviewed	2 Mar 2025 – 29 Apr 2026 (IST)	Feedback-loop timing
verdict	text	Reviewer conclusion	GENUINE 14,532 · FRAUD 4,844	New SFT labels; shadow promotion check
corrected_explanation	text	Reviewer's preferred wording	63 template sentences	New SFT / DPO data
reviewer_id	text	Which reviewer	46 synthetic reviewers	Reviewer-consistency checks

One example row (investigator review of TX_0000100016)

Column	Example value	What it tells you
feedback_id	FB_TX_0000100016	Review ID, built from the payment ID

Column	Example value	What it tells you
transaction_id	TX_0000100016	The payment reviewed
reviewed_at	2025-03-01T18:40:47.105Z	Reviewed on 2 Mar 2025 (IST), the day the label became available
verdict	FRAUD	The analyst confirmed fraud
corrected_explanation	"Unusual observed city or customer to merchant distance above 500 km; Authorisation requested between midnight and 05:00 IST; …"	The analyst's preferred wording
reviewer_id	SYNTHETIC_REVIEWER_12	Which of the 46 reviewers
is_synthetic	true	Fictional record

2.13 legacy_cases (2,000 rows) — archived 2021–2024 investigation cases

What it is: 2,000 older investigation cases from 2021 to 2024, before the main data starts, with the outcome but no rebuilt payment history. Useful as background and extra text, not for training the real-time scorer.

Attribute	Type	Business meaning	Observed values	Use in project
legacy_case_id	text, PK	Archived case ID	LEGACY_0000001…	Links to investigator notes
customer_id	text, FK	Customer in the case	2,000 distinct	Link to customers
transaction_time	timestamptz	When the old payment happened	Jan 2021 – Apr 2024	Context for CPT text
closed_at	timestamptz	When the case closed	after transaction_time	Case-handling time
amount_paise	bigint	Amount	₹59 to ₹29,984	Context
payment_channel	text	Channel	500 each of UPI, card, wallet, net_banking	Context
fraud_label	smallint	Case outcome	fraud 535 (27%) · genuine 1,465	Context; not a training label for the scorer

Attribute	Type	Business meaning	Observed values	Use in project
evidence	JSON	Saved reason codes and source	2 patterns; `history_not_reconstructed: true`	Shows old cases lack full history

One example row

Column	Example value	What it tells you
legacy_case_id	LEGACY_0000001	Archived case ID
customer_id	C_ef13d0be3f91f70c	Customer in the case
transaction_time	2024-02-24T18:30:00Z	Payment on 25 Feb 2024 (IST)
closed_at	2024-04-24T18:30:00Z	Case closed two months later
amount_paise	1983361	₹19,833.61
payment_channel	UPI	Paid by UPI
fraud_label	0	Genuine
evidence	{"reason_codes":[],"source":"synthetic_archived_snapshot","history_not_reconstructed":true}	No reason codes; the customer's earlier history was not rebuilt
is_synthetic	true	Fictional record

2.14 documents (39,006 rows) and document_chunks (78,007 rows) — the text corpus

What it is: Written text: investigator notes, chargeback stories, playbooks and regulatory notes, many tied to a payment; document_chunks splits each one into pieces of about 100 words. Used to teach the model the language of fraud (CPT) and for search and retrieval.

Attribute	Type	Business meaning	Observed values	Use in project
documents.document_id	text, PK	Document ID	e.g. DOC_investigator_notes_0023126	Join key
documents.kind	text	Document type	investigator_notes 25,834 · chargeback_narratives 11,667 · regulatory_exercises 900 · playbooks 600 · guidelines 4 ·	CPT corpus mix; RAG source types

Attribute	Type	Business meaning	Observed values	Use in project
			rba_guides 1	
documents.title	text	Title	unique per document	Search / display
documents.transaction_id	text, nullable	Related payment	10,879 null (playbooks, legacy notes)	Link note → payment
documents.legacy_case_id	text, nullable	Related archived case	29,632 null	Link note → legacy case
documents.split	text	Time split	train 30,879 · validation 4,219 · holdout 3,908	Keeps future notes out of training
documents.published_at	timestamptz	When the note was written	Mar 2021 – Apr 2026	Excludes notes written after a training cut-off
documents.content	text	Full Markdown text	~50–450 words	CPT and RAG text
documents.content_sha256	text	Fingerprint of the text	39,006 unique → 0 exact duplicates	Exact deduplication (DR-03)
documents.template_family	text	Template the text came from	505 families	Near-duplicate awareness
documents.cpt_eligible	boolean	Allowed in CPT	true 26,864 · false 12,142	Leakage control
documents.review_status	text	Human review state	always `synthetic_candidate_not_human_reviewed`	Governance
documents.storage_path / record_in_shard	text / integer	Where the text sits in the corpus files	18 JSONL shards	Locate the source
documents.metadata	JSON	Provenance (synthetic, not official, not reviewed)	—	Governance
document_chunks.c	text, PK	Chunk ID	78,007	RAG retrieval

Attribute	Type	Business meaning	Observed values	Use in project
hunk_id		(_C000, _C001)		unit
document_chunks.document_id	text, FK	Parent document	2 chunks per document (5 have 1)	Join key
document_chunks.chunk_index	integer	Order within document	0 or 1	Reassembly
document_chunks.content	text	Chunk text	53–220 words	Embedding / search input
document_chunks.word_count	integer	Length	median 99	Context-size planning
document_chunks.split / cpt_eligible	text / boolean	As documents	—	Leakage control

One example row (the investigator note on TX_0000100016 and its first chunk)

Column	Example value	What it tells you
documents.document_id	DOC_investigator_notes_0006459	Document ID
documents.kind	investigator_notes	An analyst's case note
documents.title	Investigation case note TX_0000100016 entry 1	Title names the payment
documents.transaction_id	TX_0000100016	Payment it is about
documents.legacy_case_id	empty	Not an archived case
documents.split	train	Training period
documents.published_at	2025-03-01T18:40:47.105Z	Written on 2 Mar 2025 (IST), after the label was known
documents.content	"# Investigation case note TX_0000100016 entry 1 … SYNTHETIC TRAINING MATERIAL … for INR 1418.66 through wallet …"	The note's full text
documents.content_sha256	dc63b949…79a7	Fingerprint used to find duplicates
documents.template_family	investigator_notes_electronics_1_0	Template it was generated from

Column	Example value	What it tells you
documents.cpt_eligible	true	Allowed into continued pre-training
documents.review_status	synthetic_candidate_not_human_reviewed	Not checked by a person
documents.storage_path / record_in_shard	…/corpus/investigator_notes/train/part-00003.jsonl.gz · 161	File and line where the full text sits
documents.metadata	{"label_used_in_document":true,"fraud_label":1,…}	The note reveals the answer: never show it to the scorer
document_chunks.chunk_id	DOC_investigator_notes_0006459_C000	First piece of the note
document_chunks.document_id	DOC_investigator_notes_0006459	Parent document
document_chunks.chunk_index	0	First of two chunks
document_chunks.content	"# Investigation case note TX_0000100016 entry 1 SYNTHETIC TRAINING MATERIAL …"	Text of this piece
document_chunks.word_count	220	Length of the piece
document_chunks.split / cpt_eligible	train · true	Same as the parent

2.15 gold_explanations (8,500) and preference_pairs (5,000) — SFT and DPO targets

What it is: gold_explanations are model answers: a payment written as text plus the ideal reason codes and explanation, used for SFT. preference_pairs give a good and a bad answer for the same payment so the model learns which to prefer (DPO).

Attribute	Type	Business meaning	Observed values	Use in project
gold.example_id / transaction_id	text	Example and payment IDs	`GOLD_TX_…`	Join key
gold.split	text	Split	train 8,000 · holdout 500 (evaluation pack)	Holdout 500 = Gate 2 explanation test
gold.prompt	text	fmt-v1 serialised transaction	see §5.3 of the LLD	Model input text
gold.target	JSON	Reason codes + explanation	59 distinct targets	SFT explanation target

Attribute	Type	Business meaning	Observed values	Use in project
		sentence		
gold.fraud_label	smallint	Label	fraud 3,400 (40%) · genuine 5,100	Fraud-enriched sample
gold.review_status	text	Review state	not human-reviewed	Must be reviewed before real use
dpo.pair_id / transaction_id	text	Pair and payment IDs	DPO_TX_…	Join key
dpo.prompt	text	fmt-v1 prompt	—	DPO input
dpo.chosen	JSON	Grounded explanation	55 distinct	Preferred answer
dpo.rejected	JSON	Explanation with an invented fact	198 distinct, e.g. "wrong OTP exactly seventeen times"	Teaches the model not to hallucinate
dpo.rejection_reason	text	Why rejected	always "unsupported fact absent from prompt"	Single failure type only
dpo.review_status	text	Review state	not human-reviewed	Governance

One example row from each (no payment has both a gold example and a preference pair, so the DPO row is a different payment)

Column	Example value	What it tells you
gold.example_id / transaction_id	GOLD_TX_0000100016 · TX_0000100016	Gold example for our payment
gold.split	train	Used for training
gold.prompt	[TXN] id=TX_0000100016 amt=1400.0 mcc=electronics channel=wallet time=00h Wed [CUST] age=62 acct_age_days=551 avg_90d=628.52 …	The payment written as text for the model; the amount is rounded
gold.target	{"reason_codes":["R02_GEO_MISMATCH","R03_ODD_HOUR","R08_HIGH_RISK_MCC"],"explanation":"Unusual observed city …"}	The answer the model should write
gold.fraud_label	1	Fraud
gold.review_status	synthetic_candidate_not_human_reviewed	Not checked by a person

Column	Example value	What it tells you
dpo.pair_id / transaction_id	DPO_TX_0000100129 · TX_0000100129	A gaming payment at 03:00 IST
dpo.prompt	[TXN] id=TX_0000100129 amt=2100.0 mcc=gaming channel=net_banking time=03h Wed … [TASK] Assess fraud risk using only the observed context.	The same text format, plus the task line
dpo.chosen	{… "explanation":"Unusual observed city or customer to merchant distance above 500 km; Authorisation requested between midnight and 05:00 IST; …"}	Good answer: uses only facts in the prompt
dpo.rejected	{… "explanation":"The customer entered the wrong OTP exactly seventeen times."}	Bad answer: invents a fact
dpo.rejection_reason	unsupported fact absent from prompt	Why it is bad
dpo.review_status	synthetic_candidate_not_human_reviewed	Not checked by a person

2.16 JSONL training and history files

What it is: Ready-to-use training files built from the tables: each payment's history as of that moment, SFT prompts with targets, plain text for CPT, prompt-format comparisons and a held-out evaluation set. They save you rebuilding prompts yourself.

File / field	Meaning	Observed values	Use in project
histories: transaction_id, as_of, split	Payment and moment of the snapshot	one record per transaction	Point-in-time history
histories: history[] (t, transaction_id, amount_paise, channel, mcc, city, device_id, transaction_time)	Up to 20 prior payments of the customer	0–20 entries	`[HIST]` lines of the prompt; Redis `cs:` record
training/sft: example_id, transaction_id, split, prompt_format, prompt	One fmt-v1 prompt per main transaction	train 714,745	SFT input
sft: fraud_label, reason_targets, reason_target_source	Targets for the fraud head and 8 reason heads	weak labels from feature rules	SFT targets
sft: explanation_target	Gold explanation when available	null for most rows	LM loss on gold JSON

File / field	Meaning	Observed values	Use in project
sft: sample_weight	Training weight	**always 1.0**	Class weighting still to be applied
sft: label_available_at, eligible_at_next_split_start	Label maturity	as labels	Strict historical training
sft: token_count, estimated_token_count, token_count_status, history_count	Prompt size	estimated 115–429 tokens (median 219); exact count null	1,024-token limit check
training/cpt: document_id, text	Deduplicated, cpt-eligible text	~6.8 M words, ~9.5 M estimated tokens	Continued pre-training
training/format_comparison: prompt_v1, prompt_v2, tokens_v1/v2, fraud_label	Same payment in both formats	tokens null (no tokenizer)	fmt-v1 vs fmt-v2 decision
evaluation/explanations	500 holdout gold cases	500	Gate 2 explanation evaluation only

One example record from each file

File / field	Example value	What it tells you
histories: transaction_id, as_of, split	TX_0000100016 · 2024-12-31T18:40:47.105Z · train	History as it stood at the moment of payment
histories: history[]	4 earlier payments, oldest TX_0000020082 (₹765.03 UPI, healthcare, Oct 2024, warm-up), newest TX_0000100013 (₹368.58 wallet, electronics, Chennai, same phone, 42 seconds earlier)	The past payments the features and prompt are built from
sft: example_id, transaction_id, split, prompt_format, prompt	TX_0000100016 · train · fmt-v1 · the same prompt text as the gold example	Input for supervised fine-tuning
sft: fraud_label, reason_targets, reason_target_source	1 · ["R02_GEO_MISMATCH","R03_ODD_HOUR","R08_HIGH_RISK_MCC"] · synthetic_feature_rules_weak_labels	Targets; the reason codes come from rules, not people
sft: explanation_target	{"reason_codes":[…],"explanation":"Unus	Filled for this payment (it also has a gold

File / field	Example value	What it tells you
	ual observed city …"}	example); null for most rows
sft: sample_weight	1.0	No extra weight for fraud
sft: label_available_at, eligible_at_next_split_start	2025-03-01T18:40:47.105Z · true	Same as labels
sft: token_count, estimated_token_count, token_count_status, history_count	null · 165 · estimate_only_requires_tokenizer_validation · 4	About 165 tokens; exact count needs the real tokenizer
training/cpt: document_id, text	DOC_playbooks_0000001 · "# Internal teaching playbook 0001 on device change …"	Plain text for continued pre-training
training/format_comparison	TX_0000100030 · prompt_v2 adds "[SUMMARY] hist_n=7 hist_cities=1" and drops the city from each history line · tokens_v1/v2 null · fraud_label 0	Two prompt layouts for the same payment, to compare
evaluation/explanations	GOLD_TX_0000959561 · holdout · gaming, 00h · target R02, R03, R04 · fraud_label 1	Held-out test of explanation quality

3. Business mapping

The dataset gives project every input the CRD, HLD and LLD call for, but as synthetic stand-ins: real company data, human-reviewed gold explanations and official RBI text are still needed before production.

Why a payments company collects this data

- **To stop fraud losses in real time**: every payment's context (amount, device, location, time, history) is needed to judge risk before money moves.
- **To prove decisions to regulators**: RBI and PCI DSS expect a record of who decided what and why (decision_audit, config_changes).
- **To learn from outcomes**: chargebacks and investigator verdicts tell the company which past decisions were wrong (labels, reviewer_feedback).
- **To keep good customers happy**: knowing usual behaviour lets the system avoid blocking genuine payments.

Requirement → data mapping

Project requirement	Where the data supports it

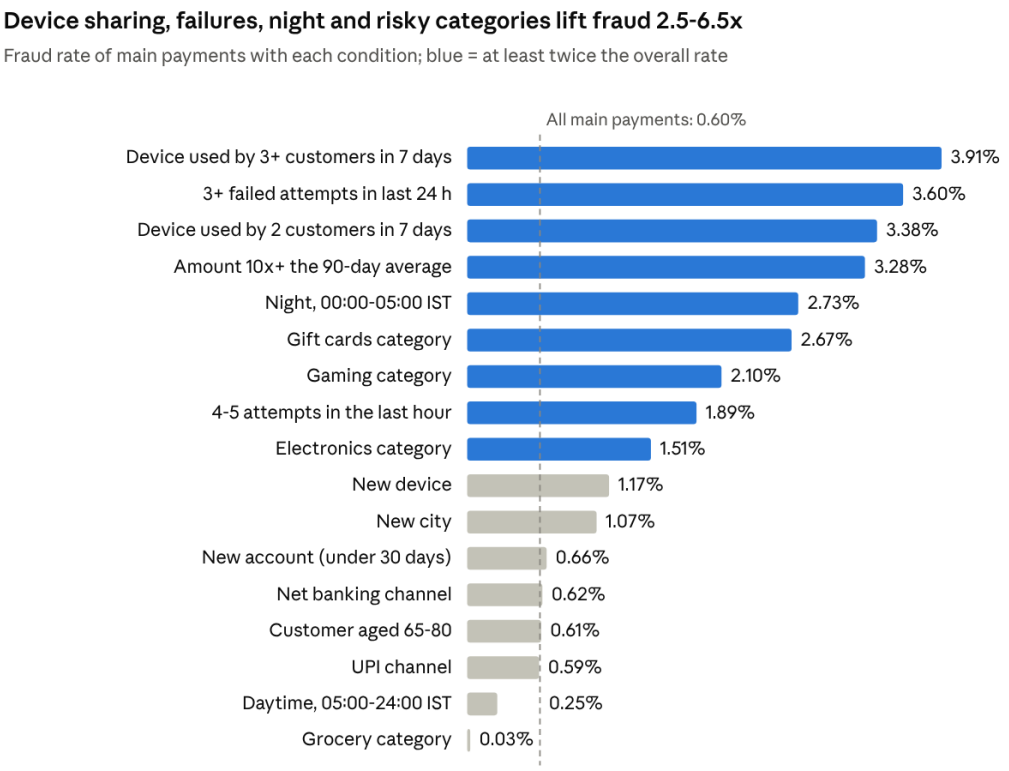
Project requirement	Where the data supports it
CRD §10.2 score request fields	transactions (all 14 CRD-required fields, including the 3 feature-store history fields)
FE-01 to FE-10 features	features table + feature_catalogue.json
Fraud label target, 60-day maturity	labels (fraud_label, label_available_at, eligible flag)
Time splits train / validation / holdout (FR-ML-05)	`split` column in every table; holdout in separate files
Class imbalance handling (FR-ML-04)	0.6% fraud rate; sample_weight field in SFT files
CPT on domain text	corpus + training/cpt (~9.5 M estimated tokens)
SFT with fraud and reason heads	training/sft (fraud_label + reason_targets)
8,000 gold explanations	gold_explanations / training/gold
DPO on reviewer preference pairs	preference_pairs / training/dpo (5,000)
Gate 2: 500 blind-rated explanations	evaluation/explanations (500)
fmt-v1 vs fmt-v2 choice (FR-FE-07)	training/format_comparison
Rules engine: thresholds, blocklist, allowlist (FR-RE-01 to 05)	threshold_sets, hard_rules, list_entries, rules_config.json
Every config change logged (FR-RE-01)	config_changes
5-year audit record (NFR-11)	decision_audit (request, features, prompt hash, versions)
Reviewer verdicts fed back (MLOPS-08)	reviewer_feedback
PII redaction and dedup (DR-01, DR-03)	content_sha256, quality_report pii_pattern_audit, cpt_eligible
Redis customer state (LLD §3.1)	histories (last 20 payments per customer)
Postgres / Aurora schema (LLD §3.2)	sql/schema.sql, indexes.sql, validate.sql

What the dataset does not cover

- Real model accuracy, latency or throughput: decisions and latencies are simulated.
- Production volume: 1 M payments over 14 months, versus 8 M a day targeted for 2028.
- Official RBI/NPCI circulars: the regulatory texts are teaching exercises only.
- Human-reviewed gold and DPO data: all marked `synthetic_candidate_not_human_reviewed`.
- AWS residency, private networking and Object Lock: the teaching environment runs on Supabase.

4. Patterns, trends and insights

Fraud concentrates in a few contexts: shared devices, recent failed attempts, night-time payments, large spikes against usual spend, and three merchant categories (gift cards, gaming, electronics). These are the attributes that matter most for project.



tables/labels, features, transactions · main splits, Jan 2025–Feb 2026 · 1,000,000 payments

Weak or no signal: payment channel, customer age band and account age all sit close to the 0.6% average. Fraud being flat across age bands is good news for the fairness check (FR-ML-11).

Other patterns and anomalies:

- **The geo flag is mostly noise.** R02_GEO_MISMATCH fires on 81% of rows because customers and merchants sit in 16 cities spread across India (median distance 1,082 km). It only roughly doubles the fraud rate, so on its own it would flood reviewers.
- **The 7 low-risk categories are almost fraud-free** (about 0.02–0.03%), so 97% of fraud is in gift cards, gaming and electronics. That supports stricter per-category thresholds (FR-RE-04).
- **Failed payments are linked to fraud**: a payment whose own outcome failed is about 10× more likely to be fraud than a successful one.
- **Large amounts are riskier**: payments above ₹50,000 are about 5× the average fraud rate, but they are only 0.4% of payments.
- **No seasonality**: the fraud rate is stable across splits (0.59%–0.64%), so drift in this data is minimal.

- **Hour 5 (05:00–05:59 IST) never occurs**, and night hours have about 60% of daytime hourly volume. Real data would not have an empty hour, so this is a generator artefact.
- **The simulated decisions catch little fraud**: the heuristic ALLOWs 89% of fraud cases (5,335 of 6,000) and only 3.7% of its BLOCKs are fraud. That is the gap a trained model has to close.
- **About four in five fraud labels come from investigator verdicts** (5,185) and one in five from confirmed chargebacks (1,415); every label arrives exactly 60 days later.

How this helps decisions

- Fraud teams can focus review capacity on the high-risk combinations above instead of on geo alerts.
- Product can set stricter thresholds for gift cards, gaming and electronics without touching low-risk categories.
- Risk can add rules for device sharing and repeated failures, which are strong, explainable signals.
- Model teams get a clear baseline to beat: the heuristic's weak recall.

5. Data quality observations

The tables are clean and consistent: no duplicate rows, no duplicate IDs, no broken links between tables, and every missing value is explained. The issues that matter are about realism, leakage and governance rather than dirty data.

Checks that passed

Check	Result
Duplicate rows (all 18 tables)	0
Duplicate primary keys	0
Orphan foreign keys (customer, merchant, device)	0
amount_paise vs transaction_amount	0 mismatches
customer_age in transactions vs customers	0 mismatches
Every transaction has an outcome, label, feature row	yes, 1:1 (1,100,000 each)
Split boundaries match the data card dates (IST)	yes
Exact duplicate documents	0 (39,006 unique SHA-256)
PII pattern scan (1,000-document sample)	0 hits, but not a human audit

Missing values (all explained)

Field	Missing	Why	Handling
device_id (and is_new_device, distinct_customers_per_device	10,683 (0.97%)	Device fingerprint not captured	unk token + missing_device flag;

Field	Missing	Why	Handling
_7d)			never impute
customer_location (and distance, is_new_city)	33,067 (3.0%)	Location not shared	`unk` + missing_customer_location flag
average_transaction_value, amount_to_avg_ratio	28,325 (2.6%)	No 90-day history (cold start)	cold_start flag
failure_code	981,435 (89%)	Only filled for FAILED payments	Expected, not a quality issue
documents.transaction_id / legacy_case_id	10,879 / 29,632	Each note links to one or the other	Expected
config_changes.before_value	1	First-ever change	Expected

Issues and anomalies to handle

1. **Extra copies on disk.** Several gzip files have been unzipped inside the folder (e.g. `tables/customers/all/part-00003.csv/part-00003.csv`, `histories/holdout/part-00036.jsonl`). Loading every file in a folder would double-count those rows. Read only the files listed in `manifest.json`.
2. **Outliers in amount.** Maximum ₹79.9 lakh against a median of ₹1,251; 317 main payments exceed ₹10 lakh. These are the "10L+" bucket in fmt-v1; for XGBoost use log(amount). amount_to_avg_ratio reaches 32,318, so cap it (fmt-v1 caps at 99.9).
3. **Hour 5 IST is empty** in all 1.1 M rows: a generator gap that a model could learn as a false pattern.
4. **R02 geo flag fires on 81% of payments**, which makes it a weak, noisy signal.
5. **Validation labels are mostly immature**: only 2% are known by 1 Jan 2026. Calibrating only on mature validation rows leaves about 2,900 examples.
6. **Fields that must never be model inputs** (leakage): fraud_probability, decision, risk_level and decision_source (simulated outputs), label_source, label_available_at, eligible flag, reviewer verdicts, ground_truth scenario, document text linked to the same payment, and every ID.
7. **SFT sample_weight is always 1.0**: the planned 1:20 down-sampling with inverse weights still has to be applied.
8. **Templated text.** 39,006 documents come from 505 template families; playbooks and regulatory exercises share near-identical bodies. Exact dedup is done; near-duplicate dedup is not certified.
9. **Token counts are estimates** (no tokenizer supplied); the largest estimated prompt is 429 tokens, under the 1,024 limit.
10. **Unused reference rows:** 166 customers have no payments and 47,102 devices are never used. Harmless, but don't count them as active.
11. **Empty and stray files in the root:** `reason code.docx` is 0 bytes (filled in §2.6 above) and `New Microsoft Excel Worksheet.xlsx` is not in the manifest.

12. **Inconsistent threshold names:** threshold_sets says medium / high / critical (0.30 / 0.60 / 0.85), while rules_config.json says medium / review / block. The data shows HIGH → REVIEW and CRITICAL → BLOCK.
13. **The Parquet transactions are not a plain copy of the CSV table.** They hold the same 1,100,000 rows with identical IDs and values (amount, age, account age, failures and history checksums all match), but drop `merchant_id` and add `fraud_label` and `label_available_at` from the labels table. Anyone training from Parquet must drop those two label columns from the inputs, or the model will see the answer.
14. **The Postgres training views leave out some inputs.** `v_train_features` in `sql/schema.sql` exposes the amount, category, age, account age, history fields, FE-01 to FE-10 and the label, but not payment_channel, raw locations, device_id or amount_paise. Channel matters little for fraud here, so this is a choice to confirm rather than an error.

Validation rules worth enforcing

- amount_paise > 0 and ≤ ₹1 crore; transaction_amount = amount_paise / 100.
- customer_age 18–120; account_age ≥ 0; failed_transactions ≥ 0.
- lat −90..90, lon −180..180; city from the known list.
- transaction_time inside its split's window; label_available_at = transaction_time + 60 days.
- fraud_label in {0, 1}; label_source = mature_no_fraud only when fraud_label = 0.
- reason_codes only from R01–R08, and only when the supporting feature flag is true.
- `sql/validate.sql` already checks 10 timing and leakage rules (money match, 60-day label delay, feature timestamps, account created before payment, outcome after payment, CPT leakage, documents after labels, DPO train-only, unknown not treated as new, frozen MCC encoder); sql/schema.sql adds CHECK constraints for amount > 0, paise = rupees × 100, account_age ≥ 0, age 18–120, valid split and fraud_label in {0, 1}; the location and reason-code checks above still need adding.

Privacy and security

- IDs are already pseudonymised hashes; no names, phones or card numbers appear in the tables.
- The text corpus passed a limited pattern scan only; a human redaction audit (DR-01) is still required for real data.
- Locations are city-level, which lowers re-identification risk, but combined with age and time they could still single people out, so access should be role-based.
- Every table in sql/schema.sql has row-level security enabled and public access revoked; holdout files must stay fenced from training roles (`sql/training_role_example.sql` shows the pattern).

6. AI/ML opportunities

The main use case is a supervised fraud classifier with explanations; the data also supports a baseline model, a RAG assistant for investigators, and monitoring. The design must use only what is known at authorisation time.

Inputs and targets

Role	Attributes
Main target	labels.fraud_label (0/1)
Secondary targets	features.reason_codes / sft.reason_targets (8 reason heads); gold target JSON (explanation text); DPO chosen vs rejected
Model inputs (raw)	amount_paise, payment_channel, merchant_category, transaction_time, customer_age, customer/merchant location, device_id presence, account_age, previous_transactions, average_transaction_value, failed_transactions, recent history
Model inputs (engineered)	FE-01 to FE-10 and the missing / cold-start flags
Never inputs	IDs, split, label metadata, simulated decisions and probabilities, reviewer verdicts, ground_truth scenarios

Preprocessing

1. Load only manifest-listed files; drop the stray unzipped copies.
2. Parse JSON columns (locations, reason codes); convert UTC to IST for hour and day features.
3. Keep the time split; for a strict backtest filter `eligible_at_next_split_start = true`.
4. Missing values: keep as `unk` plus flags, never impute.
5. Scale or bucket skewed numbers: log(amount), cap amount ratio at 99.9, distance bands as in fmt-v1.
6. Encode categories: one-hot for channel; the frozen MCC target encoding (FE-08) for category.
7. Class imbalance: weighted loss (XGBoost `scale_pos_weight` ≈ 165) or 1:20 down-sampling with inverse weights for SFT.
8. Serialise to fmt-v1 / fmt-v2 prompts for the LLM.

Feature engineering ideas beyond FE-01 to FE-10

- Interactions: night × high-risk category; failures × velocity; new device × large amount.
- Customer-relative features: amount z-score against the customer's own history; first time at this merchant or category.
- Time since the customer's last payment, and since the last failed attempt.
- Device risk over a longer window (customers per device in 30 days).
- Merchant-level velocity (sudden spikes in a merchant's volume).
- Down-weighting R02 or using raw distance instead of the noisy geo flag.

Relevant approaches

Approach	Use	Why it fits
Logistic regression / rules	Sanity baseline	Explainable; shows how much the flags alone achieve

Approach	Use	Why it fits
XGBoost / LightGBM	FR-ML-01 baseline	Strong on tabular features, fast, the bar the LLM must beat by +0.03 PR-AUC
Small LLM with classification head (SFT → DPO → distil)	Project scorer and explainer	Reads history as text; produces reason codes and explanations
Calibration (temperature / isotonic)	Turn scores into probabilities	Thresholds 0.30 / 0.60 / 0.85 need calibrated scores
Anomaly detection (isolation forest)	Unknown fraud patterns	Labels are rare and delayed
RAG over document_chunks	Investigator assistant	78,007 chunks ready for embedding and search
Drift monitoring (PSI)	MLOPS-02	Compare live distributions with training bins

Evaluation: use PR-AUC (baseline = 0.6%), recall and precision at the BLOCK threshold, rupee-weighted recall, calibration error and fairness by age band and region, all on the holdout.

Real-time vs batch

Mode	What the data supports
Real-time (per payment)	transactions = the API request; histories = the Redis state; features computed point-in-time; rules tables loaded in memory; decision_audit = the record written after each decision
Near real-time	payment_outcomes stream updates failure and velocity counters
Batch	Model training on splits; daily drift and shadow comparisons; weekly reports once labels mature (60 days); monthly retraining with reviewer_feedback

7. Discussion notes

The data is ready for exploration and baseline modelling, but its synthetic nature limits what any results can prove; we record our observations, questions and decisions here.

Key observations

- One clean, well-linked relational model centred on `transactions`, with point-in-time features already computed.
- Fraud is rare (0.6%) and concentrated in a few contexts; most risk signals are explainable.
- Leakage controls are built in (frozen MCC encoder, eligibility flag, fenced holdout, CPT cut-off), and must be kept when we load the data.

- Text and training files are synthetic templates; they are good for learning the pipeline, not for judging model quality.

Open questions for the instructor or client

1. What should `reason code.docx` contain, and was another file meant to be shared?
2. The Parquet transactions carry fraud_label and label_available_at but no merchant_id: is that intended, and which format is the source of truth for analysis?
3. Should we model with the strict `eligible_at_next_split_start` filter or the retrospective split?
4. Is the missing 05:00 IST hour intentional?
5. Should R02 geo-mismatch stay a reason code given it fires on 81% of payments?
6. Which threshold names are official: medium / high / critical or medium / review / block?
7. Will real (human-reviewed) gold and DPO data replace the synthetic candidates before training?

Decisions we propose

- Read files via `manifest.json` only, and remove or ignore the unzipped copies.
- Use the CSV tables (or Parquet) with pandas or DuckDB; avoid Excel for the large tables.
- Treat all simulated decision fields and ground_truth as evaluation-only.
- Evaluate with PR-AUC on the holdout, with the 0.6% fraud rate as the random baseline.
- Build an XGBoost baseline on FE-01 to FE-10 before any LLM work.

Learnings

- Understanding *when* each value becomes known matters as much as what it means: the 60-day label delay and point-in-time features shape the whole design.
- A high-frequency flag (R02) can look useful but carry little signal; rates by segment reveal this quickly.
- A simulated decision table is a useful template for the audit design but must never leak into training.

8. Individual contributions

Replace each member placeholder with a name; the split below follows the report's sections so each person can explain their part.

Member	Explored and explained	Report sections
Member 1	Metadata files (data card, manifest, quality report, requirements coverage), folder structure, record counts	1, 3
Member 2	customers, merchants, devices, transactions	2.1–2.4
Member 3	features, reason codes, labels, payment outcomes, ground truth; fraud-rate patterns	2.5–2.9, 4
Member 4	decision audit, rules and configuration, reviewer feedback; documents, chunks, training and history files	2.10–2.16

Member	Explored and explained	Report sections
Member 5	Data quality checks, validation rules, privacy; AI/ML opportunities	5, 6
All	Discussion notes, open questions, decisions	7